Product Description

Product Information	
Product name	Picosecond Pulsed Fiber-Coupled Laser
Product number	PLD-XXX-SM-FC-U
Manufacturer	FLIM LABS S.r.l.
Registered Address	Via della Farnesina, 3 – 00135 Rome (RM), Italy



General Description

The PLD is a solid-state diode laser source that can operate in both continuous (CW) and pulsed modes with selectable repetition frequencies and output power settings. The laser emits at a single wavelength selected at the time of purchase from the available models listed in Table 1. Depending on the selected wavelength, the emitted radiation is visible or invisible.

The accessible laser emission is limited such that the product is classified as a Class 3R laser product in accordance with IEC EN CEI 60825-1 ed.3 (2014).

The source is a self-contained unit designed to operate on a desktop or laboratory bench and to be connected to a microscopy instrument or other imaging equipment via an optical fiber.

Table 1: Available PLD laser models.

Model	Wavelength	Radiation Type
PLD-405-SM-FC-U	405 nm	Visible
PLD-445-SM-FC-U	445 nm	Visible
PLD-488-SM-FC-U	488 nm	Visible
PLD-520-SM-FC-U	520 nm	Visible
PLD-635-SM-FC-U	635 nm	Visible
PLD-785-SM-FC-U	785 nm	Invisible

Model	Wavelength	Radiation Type
PLD-808-SM-FC-U	808 nm	Invisible
PLD-850-SM-FC-U	850 nm	Invisible

The model number encodes the product specifications in the following scheme:

- SM stands for single-mode
- FC for fiber-receptacle
- U indicates the USB power interface

The device incorporates a Class 3B laser diode with a control system that limits the laser diode power such that, including under fault conditions, the accessible laser output remains within the limits applicable to a Class 3R laser product.
The laser output is coupled into a single-mode optical fiber with a 4 µm core, providing a near-Gaussian spatial beam profile.

Intended Use and Misuse

Intended Use

The device described in this manual is designed for standalone use or incorporation into other equipment or machinery for research and diagnostic applications in the field of fluorescence analysis.

Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 10°Cto +40°C
- Storage temperature: −20°Cto +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases. Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.